

Exercise #8.3

Find an equation of the plane through the three given points:

Q#1: $(2, 1, 1), (6, 3, 1), (-2, 1, 2)$.

Solution:

Given points are $P(2,1,1), Q(6,3,1)$ & $R(-2,1,2)$

Let required equation of the plane is

$$ax + by + cz + d = 0 \quad \text{----- (1)}$$

Since it passes through the points $P(2,1,1), Q(6,3,1)$ & $R(-2,1,2)$

$$\text{So } 2a + b + c + d = 0 \quad \text{----- (2)}$$

$$6a + 3b + c + d = 0 \quad \text{----- (3)}$$

$$-2a + b + 2c + d = 0 \quad \text{----- (4)}$$

Subtracting eq. (2), (3) & (3), (4)

$$\Rightarrow -4a - 2b + 0c = 0$$

$$\begin{aligned} 2a + b + 0c &= 0 \\ 8a + 2b - c &= 0 \end{aligned}$$

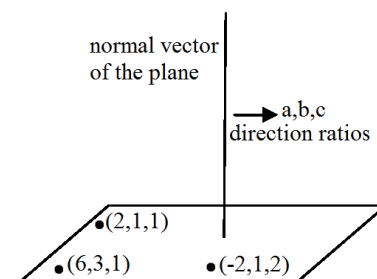
$$\frac{a}{\begin{vmatrix} 1 & 0 \\ 2 & -1 \end{vmatrix}} = \frac{-b}{\begin{vmatrix} 2 & 0 \\ 8 & -1 \end{vmatrix}} = \frac{c}{\begin{vmatrix} 2 & 1 \\ 8 & 2 \end{vmatrix}} \Rightarrow \frac{a}{-1-0} = \frac{-b}{-2-0} = \frac{c}{4-8} \Rightarrow \frac{a}{-1} = \frac{b}{2} = \frac{c}{-4} \Rightarrow a = -1, b = 2 \text{ \& } c = -4$$

Putting these values in equation (2)

$$2(-1) + 2 + (-4) + d = 0 \Rightarrow -2 + 2 - 4 + d = 0 \Rightarrow d = 4$$

Now putting all values in equation (1) $(-1)x + 2y + (-4)z + 4 = 0 \Rightarrow -x + 2y - 4z + 4 = 0$

$$x - 2y + 4z - 4 = 0 \text{ is required equation of plane}$$



Method II.

Given points are $P(2,1,1), Q(6,3,1)$ & $R(-2,1,2)$

Let required equation of the plane is

$$ax + by + cz + d = 0 \quad \text{----- (1)}$$

$$\text{For point } P(2,1,1) \Rightarrow 2a + b + c + d = 0 \quad \text{----- (2)}$$

$$\text{For point } Q(6,3,1) \Rightarrow 6a + 3b + c + d = 0 \quad \text{----- (3)}$$

$$\text{For point } R(-2,1,2) \Rightarrow -2a + b + 2c + d = 0 \quad \text{----- (4)}$$

Eliminating a, b, c and d from equation (1), (2), (3) & (4)

we get an equation of the required plane as

$$\begin{vmatrix} x & y & z & 1 \\ 2 & 1 & 1 & 1 \\ 6 & 3 & 1 & 1 \\ -2 & 1 & 2 & 1 \end{vmatrix} = 0$$

By using row operations

$$\begin{vmatrix} x+2 & y-1 & z-2 & 0 \\ 4 & 0 & -1 & 0 \\ 8 & 2 & -1 & 0 \\ -2 & 1 & 2 & 1 \end{vmatrix} = 0,$$

$$\text{by } R_1 - R_4, R_2 - R_4 \\ \text{and } R_3 - R_4$$

Now expanding by C_4

$$\begin{vmatrix} x+2 & y-1 & z-2 \\ 4 & 0 & -1 \\ 8 & 2 & -1 \end{vmatrix} = 0$$

$$-4 \begin{vmatrix} y-1 & z-2 \\ 2 & -1 \end{vmatrix} + 1 \begin{vmatrix} x+2 & y-1 \\ 8 & 2 \end{vmatrix} = 0$$

$$\Rightarrow -4(-1(y-1) - 2(z-2)) + 1(2(x+2) - 8(y-1)) = 0$$

$$\Rightarrow -4(-y+1-2z+4) + 1(2x+4-8y+8) = 0$$

$$\Rightarrow 4y-4+8z-16+2x+4-8y+8 = 0$$

$$\Rightarrow 2x-4y+8z-8 = 0$$

$$\Rightarrow x-2y+4z-4 = 0 \quad \text{is required equation of plane.}$$

Q#2: (1, -1, 2), (-3, -2, 6), (6, 0, 1). DO YOURSELF AS ABOVE

Q#3: (-1, 1, 1), (2, -8, -2), (4, 1, 0). DO YOURSELF AS ABOVE

Q#4: Find equations of the plane bisecting the angles between the planes

$3x + 2y - 6z + 1 = 0$ and $2x + y + 2z - 5 = 0$.

Solution: Given equations of the plane are

$$3x + 2y - 6z + 1 = 0 \quad \text{-----} (P_1)$$

$$2x + y + 2z - 5 = 0 \quad \text{-----} (P_2)$$

Let a point A on the plane (P_1) and point B on the plane (P_2) and point $P(x_1, y_1, z_1)$ on the required plane bisecting the angles between given planes.

Then the distance of point $P(x_1, y_1, z_1)$ from both planes should be equal.

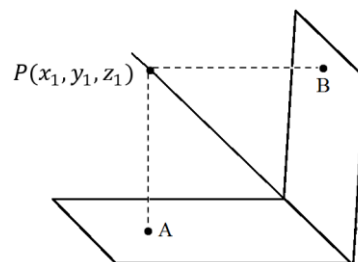
From the figure $|\vec{AP}| = |\vec{BP}|$

$$\frac{|3x_1 + 2y_1 - 6z_1 + 1|}{\sqrt{3^2 + 2^2 + 6^2}} = \frac{|2x_1 + y_1 + 2z_1 - 5|}{\sqrt{2^2 + 1^2 + 2^2}}$$

$$\frac{3x_1 + 2y_1 - 6z_1 + 1}{7} = \pm \frac{2x_1 + y_1 + 2z_1 - 5}{3}$$

$$\frac{3x_1 + 2y_1 - 6z_1 + 1}{7} = \frac{2x_1 + y_1 + 2z_1 - 5}{3}$$

$$\begin{aligned} \Rightarrow 3(3x_1 + 2y_1 - 6z_1 + 1) &= 7(2x_1 + y_1 + 2z_1 - 5) \\ \Rightarrow 9x_1 + 6y_1 - 18z_1 + 3 &= 14x_1 + 7y_1 + 14z_1 - 35 \\ 9x_1 + 6y_1 - 18z_1 + 3 - 14x_1 - 7y_1 - 14z_1 + 35 &= 0 \\ \Rightarrow -5x_1 - y_1 - 32z_1 + 38 &= 0 \\ 5x_1 + y_1 + 32z_1 - 38 &= 0 \end{aligned}$$



$$\frac{3x_1 + 2y_1 - 6z_1 + 1}{7} = -\frac{2x_1 + y_1 + 2z_1 - 5}{3}$$

$$\begin{aligned} \Rightarrow 3(3x_1 + 2y_1 - 6z_1 + 1) &= -7(2x_1 + y_1 + 2z_1 - 5) \\ \Rightarrow 9x_1 + 6y_1 - 18z_1 + 3 &= -14x_1 - 7y_1 - 14z_1 + 35 \\ 9x_1 + 6y_1 - 18z_1 + 3 + 14x_1 + 7y_1 + 14z_1 - 35 &= 0 \\ 23x_1 + 13y_1 - 4z_1 - 32 &= 0 \end{aligned}$$

Q#5: Transform the equations of the planes $3x - 4y + 5z = 0$ and $2x - y - 2z = 5$ to normal form and hence find measure of the angle between them.

Solution: Given equations of the plane are

$$3x - 4y + 5z = 0 \quad \text{-----} (P_1) \quad \& \quad 2x - y - 2z = 5 \quad \text{-----} (P_2)$$

Let $\vec{n}_1 = 3\hat{i} - 4\hat{j} + 5\hat{k}$ is a normal vector of plane (P_1)

& $\vec{n}_2 = 2\hat{i} - \hat{j} - 2\hat{k}$ is a normal vector of plane (P_2)

$$|\vec{n}_1| = \sqrt{3^2 + 4^2 + 5^2} = \sqrt{50} \quad (P_1) \Rightarrow \quad \frac{3}{\sqrt{50}}x - \frac{4}{\sqrt{50}}y + \frac{5}{\sqrt{50}}z = 0$$

$$|\vec{n}_2| = \sqrt{2^2 + 1^2 + 2^2} = \sqrt{9} = 3 \quad (P_2) \Rightarrow \quad \frac{2}{3}x - \frac{1}{3}y - \frac{2}{3}z = \frac{5}{3}$$

Let θ be the angle between (P_1) & (P_2) then by

$$\cos \theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1||\vec{n}_2|} = \frac{(3\hat{i} - 4\hat{j} + 5\hat{k}) \cdot (2\hat{i} - \hat{j} - 2\hat{k})}{(3)\sqrt{50}} = \frac{6 + 4 - 10}{3\sqrt{50}} = \frac{10 - 10}{3\sqrt{50}} = 0 \Rightarrow \theta = \cos^{-1}(0) \Rightarrow \theta = 90^\circ$$

Q#6: Find equations to the plane through the points $(4, -5, 3)$, $(2, 3, 1)$ and parallel to the coordinate axis.

Solution: Let required equation of the plane is

$$ax + by + cz + d = 0 \quad \text{-----} (1)$$

Since it passes through the points $(4, -5, 3)$, $(2, 3, 1)$

$$\text{So} \quad 4a - 5b + 3c + d = 0 \quad \text{-----} (2)$$

$$2a + 3b + c + d = 0 \quad \text{-----} (3)$$

Subtracting eq. (2) & (3)

$$2a - 8b + 2c = 0 \quad \text{-----} (4)$$

Case (I): Since required plane is parallel to x- axis whose direction ratios are 1,0,0

$$\text{So (1)} \Rightarrow a + 0b + 0c = 0 \quad \text{-----} (5)$$

Now using eq. (4) & (5)

$$\frac{a}{\begin{vmatrix} -8 & 2 \\ 0 & 0 \end{vmatrix}} = \frac{-b}{\begin{vmatrix} 2 & 2 \\ 1 & 0 \end{vmatrix}} = \frac{c}{\begin{vmatrix} 2 & -8 \\ 1 & 0 \end{vmatrix}} \Rightarrow \frac{a}{0-0} = \frac{-b}{0-2} = \frac{c}{0+8} \Rightarrow \frac{a}{0} = \frac{b}{2} = \frac{c}{8} \Rightarrow \frac{a}{0} = \frac{b}{1} = \frac{c}{4}$$

Putting these values in eq. (3)

$$0 + 3 + 4 + d = 0 \Rightarrow d = -7$$

Put all values in eq. (1)

$$0x + y + 4z - 7 = 0$$

$$\Rightarrow y + 4z - 7 = 0 \quad \text{is required equation of plane}$$

Case (II): Since required plane is parallel to y- axis whose direction ratios are 0,1,0

$$\text{So (1)} \Rightarrow 0a + b + 0c = 0 \quad \text{-----} (6)$$

Now using eq. (4) & (6)

$$\frac{a}{\begin{vmatrix} -8 & 2 \\ 1 & 0 \end{vmatrix}} = \frac{-b}{\begin{vmatrix} 2 & 2 \\ 0 & 0 \end{vmatrix}} = \frac{c}{\begin{vmatrix} 2 & -8 \\ 0 & 1 \end{vmatrix}} \Rightarrow \frac{a}{0-2} = \frac{-b}{0-0} = \frac{c}{2-0} \Rightarrow \frac{a}{-2} = \frac{b}{0} = \frac{c}{2} \Rightarrow \frac{a}{1} = \frac{b}{0} = \frac{c}{-1}$$

Putting these values in eq. (3)

$$2(1) + 0 - 1 + d = 0 \Rightarrow d = -1$$

Put all values in eq. (1)

$$x + 0y - z - 1 = 0 \Rightarrow x - z - 1 = 0 \text{ is required equation of plane}$$

Case (III): Since required plane is parallel to z- axis whose direction ratios are 0,0,1

$$\text{So (1)} \Rightarrow 0a + 0b + c = 0 \text{ ----- (7)}$$

Now using eq. (4) & (7)

$$\frac{a}{\begin{vmatrix} -8 & 2 \\ 0 & 1 \end{vmatrix}} = \frac{-b}{\begin{vmatrix} 2 & 2 \\ 0 & 1 \end{vmatrix}} = \frac{c}{\begin{vmatrix} 2 & -8 \\ 0 & 0 \end{vmatrix}} \Rightarrow \frac{a}{-8-0} = \frac{-b}{2-0} = \frac{c}{0-0} \Rightarrow \frac{a}{-8} = \frac{-b}{2} = \frac{c}{0} \Rightarrow \frac{a}{4} = \frac{b}{1} = \frac{c}{0}$$

Putting these values in eq. (3)

$$2(4) + 3 + 0 + d = 0 \Rightarrow d = -11$$

Put all values in eq. (1)

$$4x + y - 0z - 11 = 0 \Rightarrow 4x + y - 11 = 0 \text{ is required equation of plane.}$$

Q#7: Find an equation of the plane through the points (1, 0, 1) and (2, 2, 1) and perpendicular to the plane $x - y - z + 4 = 0$.

Solution: Let required equation of the plane is

$$ax + by + cz + d = 0 \text{ ----- (1)}$$

It passes through the points (1,0,1) and (2,2,1)

$$(1,0,1) \Rightarrow a + 0b + c + d = 0 \text{ ----- (2)}$$

$$(2,2,1) \Rightarrow 2a + 2b + c + d = 0 \text{ ----- (3)}$$

Subtracting eq. (2) & (3)

$$-a - 2b + 0c = 0 \text{ ----- (4)}$$

Given equation of the plane is

$$x - y - z + 4 = 0 \text{ ----- (5)}$$

$$\text{Let for plane (1)} \quad \vec{n}_1 = a\hat{i} + b\hat{j} + c\hat{k}$$

$$\text{Let for plane (5)} \quad \vec{n}_2 = \hat{i} - \hat{j} - \hat{k}$$

As plane (1) is perpendicular to given plane . So by condition of perpendicularity

$$\vec{n}_1 \perp \vec{n}_2 \Rightarrow \vec{n}_1 \cdot \vec{n}_2 = 0$$

$$(a\hat{i} + b\hat{j} + c\hat{k}) \cdot (\hat{i} - \hat{j} - \hat{k}) = 0$$

$$a - b - c = 0 \text{ ----- (6)}$$

Using eq. (4) & (6)

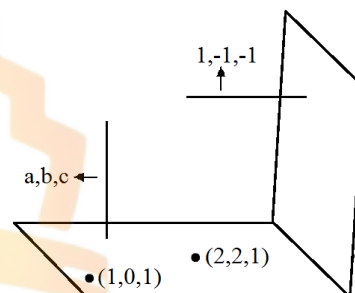
$$\frac{a}{\begin{vmatrix} -2 & 0 \\ -1 & -1 \end{vmatrix}} = \frac{-b}{\begin{vmatrix} -1 & 0 \\ 1 & -1 \end{vmatrix}} = \frac{c}{\begin{vmatrix} -1 & -2 \\ 1 & -1 \end{vmatrix}} \Rightarrow \frac{a}{2+0} = \frac{-b}{1-0} = \frac{c}{1+2} \Rightarrow \frac{a}{2} = \frac{b}{-1} = \frac{c}{3}$$

Putting these proportional values of a ,b & c in eq. (2)

$$2 + 0 + 3 + d = 0 \Rightarrow d = -5$$

Now putting all values in equation (1)

$$2x - y + 3z - 5 = 0 \text{ required equation of plane.}$$



Q#8: Find an equation of the plane which is perpendicular bisector of the line segment joining the points $(3, 4, -1)$ and $(5, 2, 7)$.

Solution: Let AB is a line segment.

Coordinates of given points are $A(3, 4, -1)$ & $B(5, 2, 7)$.

Now direction ratios of line AB are $\overrightarrow{AB} = B(5, 2, 7) - A(3, 4, -1)$

$\overrightarrow{AB} = 2\hat{i} - 2\hat{j} + 8\hat{k}$ it is also normal vector of the required plane.

As line $\overrightarrow{AB}$ is perpendicular to required plane so midpoint M of line $\overrightarrow{AB}$ is

$$M = \left(\frac{3+5}{2}, \frac{4+2}{2}, \frac{-1+7}{2} \right) = (4, 3, 3)$$

As plane is perpendicular bisector of line $\overrightarrow{AB}$ so point $M = (4, 3, 3)$ lies on required plane

Hence required equation of the

plane through the point $M = (4, 3, 3)$

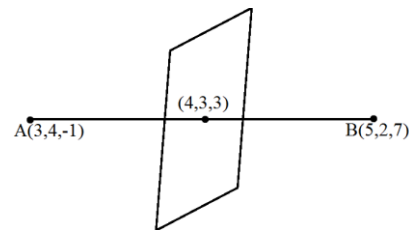
having normal vector $\overrightarrow{AB} = 2\hat{i} - 2\hat{j} + 8\hat{k}$.

$$2(x - 4) - 2(y - 3) + 8(z - 3) = 0$$

$$2x - 2y + 8z - 8 + 6 - 24 = 0$$

$$2x - 2y + 8z - 26 = 0$$

$$x - y + 4z - 13 = 0 \quad \text{required plane.}$$



Q#9: Show that the join of $(0, -1, 0)$ and $(2, 4, -1)$ intersects the join of $(1, 1, 1)$ and $(3, 3, 9)$.

Solution: First we will show that the four given points are coplanar.

Now we find equation of the plane through three points.

Let the equation of required plane is

$$ax + by + cz + d = 0 \quad \text{----- (1)}$$

As it passes through $(0, -1, 0)$, $(2, 4, -1)$ & $(1, 1, 1)$

$$\text{So } 0a - b + 0c + d = 0 \quad \text{----- (2)}$$

$$2a + 4b - c + d = 0 \quad \text{----- (3)}$$

$$a + b + c + d = 0 \quad \text{----- (4)}$$

Subtracting eq. (3) from (2) & eq. (4) from (3)

$$\Rightarrow \begin{cases} -2a - 5b + c = 0 \\ a + 3b - 2c = 0 \end{cases}$$

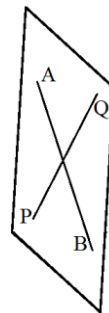
$$\frac{a}{\begin{vmatrix} -5 & 1 \\ 3 & -2 \end{vmatrix}} = \frac{-b}{\begin{vmatrix} -2 & 1 \\ 1 & -2 \end{vmatrix}} = \frac{c}{\begin{vmatrix} -2 & -5 \\ 1 & 3 \end{vmatrix}} \Rightarrow \frac{a}{10-3} = \frac{-b}{4-1} = \frac{c}{-6+5} \Rightarrow \frac{a}{7} = \frac{b}{-3} = \frac{c}{-1}$$

Now putting these proportional values of a, b & c in eq. (2)

$$0 + 3 + 0 + d = 0 \Rightarrow d = -3$$

Put all values in eq. (1)

$$7x - 3y - z - 3 = 0$$



Put the fourth point (3,3,9) in above equation.

$$7(3) - 3(3) - 9 - 3 = 0$$

$$21 - 9 - 9 - 3 = 0$$

$$21 - 21 = 0 \quad \Rightarrow 0 = 0$$

As the equation of plane is satisfied, hence the four points are coplanar. Hence the two joins are coplanar.

Now direction ratios of join of (0, -1, 0) and (2, 4, -1) are $2 - 0, 4 + 1, -1 - 0 \Rightarrow 2, 5, -1$

& direction ratios of join of (1, 1, 1) and (3, 3, 9) are $3 - 1, 3 - 1, 9 - 1 \Rightarrow 2, 2, 8$

As direction ratios of both joins are not proportional so the two joins are not parallel & so being coplanar they intersect each other.

Q#10: The vertices of tetrahedron are (0, 0, 0), (3, 0, 0), (0, -4, 0) and (0, 0, 5). Find equations of planes of its faces.

Solution: Let the vertices of given tetrahedron are $A(0,0,0), B(3,0,0), C(0,-4,0)$ and $D(0,0,5)$.

Then we want to find the equations of the plane faces ABC, ABD, ACD & BCD .

(I) Equation of plane for face ABC

Let the equation of required plane is

$$ax + by + cz + d = 0 \quad \text{----- (1)}$$

As it passes through $A(0,0,0), B(3,0,0)$ & $C(0,-4,0)$

$$\text{So } 0a + 0b + 0c + d = 0 \quad \text{----- (2)}$$

$$3a + 0b + 0c + d = 0 \quad \text{----- (3)}$$

$$0a - 4b + 0c + d = 0 \quad \text{----- (4)}$$

Subtracting eq. (3) from (2) and (4) from (3)

$$\begin{aligned} -3a + 0b + 0c &= 0 \\ 3a + 4b + 0c &= 0 \end{aligned}$$

$$\frac{a}{0-0} = \frac{-b}{0-0} = \frac{c}{-12-0} \Rightarrow \frac{a}{0} = \frac{b}{0} = \frac{c}{-12} \Rightarrow \frac{a}{0} = \frac{b}{0} = \frac{c}{1}$$

Putting these proportional values of a, b & c in eq. (2)

$$0 + 0 + 0 + d = 0 \Rightarrow d = 0$$

Put all values in eq. (1)

$$0x + 0y + z + 0 = 0 \Rightarrow z = 0$$

(II) Equation of plane for face ABD

Let the equation of required plane is

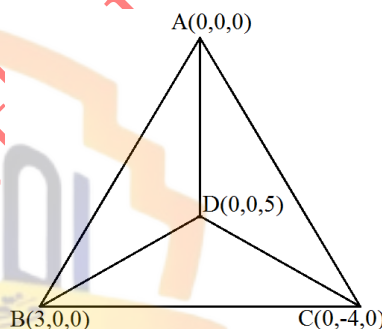
$$ax + by + cz + d = 0 \quad \text{----- (1)}$$

As it passes through $A(0,0,0), B(3,0,0)$ & $D(0,0,5)$

$$\text{So } 0a + 0b + 0c + d = 0 \quad \text{----- (2)}$$

$$3a + 0b + 0c + d = 0 \quad \text{----- (3)}$$

$$0a - 0b + 5c + d = 0 \quad \text{----- (4)}$$



Subtracting eq. (3) from (2) and (4) from (3)

$$\begin{cases} -3a + 0b + 0c = 0 \\ 3a + 0b - 5c = 0 \end{cases}$$

$$\frac{a}{0-0} = \frac{-b}{15-0} = \frac{c}{0-0} \Rightarrow \frac{a}{0} = \frac{b}{-15} = \frac{c}{0} \Rightarrow \frac{a}{0} = \frac{b}{1} = \frac{c}{0}$$

Putting these proportional values of a, b & c in eq. (2)

$$0 + 0 + 0 + d = 0 \Rightarrow d = 0$$

Put all values in eq. (1)

$$0x + y + 0z + 0 = 0$$

$$y = 0$$

(III) Equation of plane face ACD & BCD DO YOURSELF AS ABOVE

Q#11: Find an equation of the plane through $(5, -1, 4)$ and perpendicular to each of the planes $x + y - 2z - 3 = 0$ and $2x - 3y + z = 0$.

Solution:

Let the equation of required plane is

$$ax + by + cz + d = 0 \text{ ----- (1)}$$

The normal vector of required plane is $\vec{n} = a\hat{i} + b\hat{j} + c\hat{k}$

As plane (1) passes through $(5, -1, 4)$, so

$$5a - b + 4c + d = 0 \text{ ----- (2)}$$

As plane (1) is perpendicular to given

planes $x + y - 2z - 3 = 0$ and $2x - 3y + z = 0$

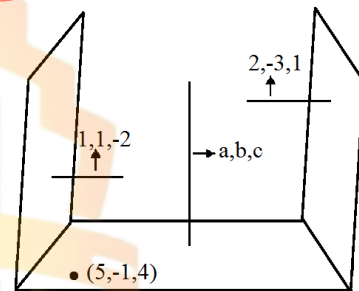
Normal vector of given plane (1) is $\vec{n}_1 = \hat{i} + \hat{j} - 2\hat{k}$

Normal vector of given plane (2) is $\vec{n}_2 = 2\hat{i} - 3\hat{j} + \hat{k}$

By using given condition

$$\begin{aligned} \vec{n} \perp \vec{n}_1 \quad \text{Then} \quad \vec{n} \cdot \vec{n}_1 &= 0 \\ (a\hat{i} + b\hat{j} + c\hat{k}) \cdot (\hat{i} + \hat{j} - 2\hat{k}) &= 0 \\ a + b - 2c &= 0 \text{ ----- (3)} \end{aligned}$$

$$\begin{aligned} \vec{n} \perp \vec{n}_2 \quad \text{Then} \quad \vec{n} \cdot \vec{n}_2 &= 0 \\ (a\hat{i} + b\hat{j} + c\hat{k}) \cdot (2\hat{i} - 3\hat{j} + \hat{k}) &= 0 \\ 2a - 3b + c &= 0 \text{ ----- (4)} \end{aligned}$$



Now using eq. (3) & (4)

$$\frac{a}{\begin{vmatrix} 1 & -2 \\ -3 & 1 \end{vmatrix}} = \frac{-b}{\begin{vmatrix} 1 & -2 \\ 2 & 1 \end{vmatrix}} = \frac{c}{\begin{vmatrix} 1 & 1 \\ 2 & -3 \end{vmatrix}} \Rightarrow \frac{a}{1-6} = \frac{-b}{1+4} = \frac{c}{-3-2} \Rightarrow \frac{a}{-5} = \frac{-b}{5} = \frac{c}{-5} \Rightarrow \frac{a}{1} = \frac{b}{1} = \frac{c}{1}$$

Using these proportional values of a, b & c in eq. (2)

$$5 - 1 + 4 + d = 0 \Rightarrow d = -8$$

Put all values in eq. (1) we get $x + y + z - 8 = 0$ is required equation of plane.

Q#12: Find an equation of the plane each of whose point is equidistant from the points $A(2, -1, 1)$ and $B(3, 1, 5)$.

Solution:

As the required plane is equidistant from the points $A(2, -1, 1)$ and $B(3, 1, 5)$,

so it should be a perpendicular bisector of the line AB & line AB

will be a normal vector of the required plane.

Now direction ratios of line AB are

$$\overrightarrow{AB} = B(3, 1, 5) - A(2, -1, 1) \Rightarrow \hat{i} + 2\hat{j} + 4\hat{k} \quad \text{it is also normal vector of the required plane}$$

$$\text{midpoint of AB is } M = \left(\frac{2+3}{2}, \frac{-1+1}{2}, \frac{1+5}{2} \right) = \left(\frac{5}{2}, 0, 3 \right)$$

Now equation of plane through the point $M\left(\frac{5}{2}, 0, 3\right)$

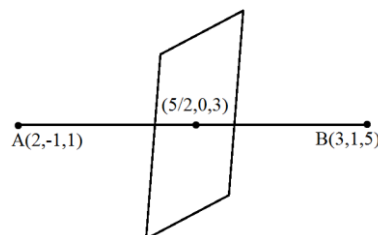
having normal with direction ratios 1, 2, 4

$$1\left(x - \frac{5}{2}\right) + 2(y - 0) + 4(z - 3) = 0$$

$$x + 2y + 4z - \frac{5}{2} - 12 = 0$$

$$2x + 4y + 8z - 5 - 24 = 0$$

$$2x + 4y + 8z - 29 = 0 \quad \text{is required equation of plane.}$$



Q#13: Find an equation of the plane through the point $(3, -2, 5)$ and perpendicular to the line

$$x = 2 + 3t, y = 1 - 6t, z = -2 + 2t.$$

Solution:

given that the required plane passes through the point $(3, -2, 5)$ & it is perpendicular to the given line

$$x = 2 + 3t, \quad y = 1 - 6t, \quad z = -2 + 2t.$$

Here direction ratios of given line are 3, -6, 2

Normal vector of the plane is parallel to the given line,

therefore direction ratios of the normal vector of the plane are

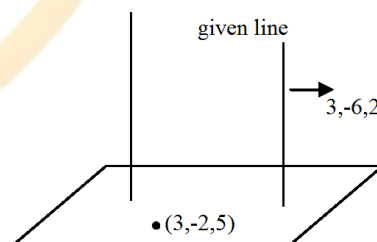
$$a = 3, \quad b = -6, \quad c = 2$$

Hence required equation of the plane through $(3, -2, 5)$.

$$3(x - 3) - 6(y + 2) + 2(z - 5) = 0$$

$$3x - 9 - 6y - 12 + 2z - 10 = 0$$

$$3x - 6y + 2z - 31 = 0 \quad \text{required equation of plane}$$



Q#14: Find parametric equations of the line containing the point $(2, 4, -3)$ and perpendicular to the plane

$$3x + 3y - 7z = 9.$$

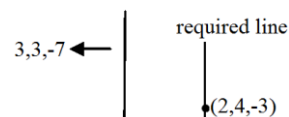
Solution:

As given that required line is perpendicular to the given plane $3x + 3y - 7z = 9$

we see that the direction ratios of given planes are 3, 3, -7.

As required line is parallel to the normal vector

of the plane so direction ratios of required line will 3, 3, -7.



Now equation of required line through $(2, 4, -3)$ having

direction ratios $3, 3, -7$.

$$\frac{x-2}{3} = \frac{y-4}{3} = \frac{z+3}{-7} = t \text{ (say)}$$

Now parametric equations of the above line are

$$\Rightarrow \begin{cases} x = 2 + 3t \\ y = 4 + 3t \\ z = -3 - 7t \end{cases}$$

Q#15: Write equation of the family of all planes whose distance from the origin is 7. Find those members of the family which are parallel to the plane $x + y + z + 5 = 0$.

Solution:

The equation of family of all planes in normal form is

$$lx + my + nz = p \text{ ----- (1)}$$

As given $p = 7$ p is the distance between plane and origin

Then above equation becomes $lx + my + nz = 7$

Here l, m & n are the direction cosines of normal vector of the plane.

Equation of given plane is

$$x + y + z + 5 = 0$$

$$\text{or } x + y + z = -5$$

Dividing both sides by $\sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$

$$\pm \frac{x}{\sqrt{3}} \pm \frac{y}{\sqrt{3}} \pm \frac{z}{\sqrt{3}} = \pm \left(-\frac{5}{\sqrt{3}}\right)$$

$$\frac{x}{\sqrt{3}} + \frac{y}{\sqrt{3}} + \frac{z}{\sqrt{3}} = -\frac{5}{\sqrt{3}} \text{ and } -\frac{x}{\sqrt{3}} - \frac{y}{\sqrt{3}} - \frac{z}{\sqrt{3}} = \frac{5}{\sqrt{3}} \text{ ----- (2)}$$

A plane parallel to (1) has normal vector with direction cosines $-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}$ or $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$

Here two sets of family of plane parallel to plane (1) so these members are

$$-\frac{1}{\sqrt{3}}x - \frac{1}{\sqrt{3}}y - \frac{1}{\sqrt{3}}z = 7 \text{ \& } \frac{1}{\sqrt{3}}x + \frac{1}{\sqrt{3}}y + \frac{1}{\sqrt{3}}z = 7$$

Q#16: Find an equation of the plane which passes through the point $(3, 4, 5)$ has an x-intercept equal to -5 and is perpendicular to the plane $2x + 3y - z = 8$.

Solution:

Equation of a plane in intercept form is

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

As given x-intercept $a = -5$

Putting in above equation

$$\frac{x}{-5} + \frac{y}{b} + \frac{z}{c} = 1 \text{ ----- (1)}$$

As this plane is perpendicular to $2x + 3y - z = 8$

So by condition of perpendicularity.

$$-\frac{1}{5}(2) + \frac{1}{b}(3) + \frac{1}{c}(-1) = 0 \Rightarrow -\frac{2}{5} + \frac{3}{b} - \frac{1}{c} = 0 \Rightarrow \frac{3}{b} - \frac{1}{c} = \frac{2}{5}$$

Also since plane (1) passes through point (3,4,5)

$$\text{so } \frac{3}{-5} + \frac{4}{b} + \frac{5}{c} = 1 \Rightarrow \frac{4}{b} + \frac{5}{c} = 1 + \frac{3}{5}$$

$$\frac{4}{b} + \frac{5}{c} = \frac{8}{5} \quad \text{--- (3)}$$

$$\frac{15}{b} - \frac{5}{c} = 2 \quad \text{--- (4)} \quad \text{multiplying eq. (2) by 5}$$

Adding eq. (3) & (5)

$$\frac{4}{b} + \frac{15}{b} = \frac{8}{5} + 2 \Rightarrow \frac{19}{b} = \frac{18}{5} \Rightarrow 18b = 95 \Rightarrow b = \frac{95}{18}$$

Putting in eq. (2)

$$\frac{3}{95/18} - \frac{1}{c} = \frac{2}{5} \Rightarrow \frac{54}{95} - \frac{1}{c} = \frac{2}{5} \Rightarrow \frac{54}{95} - \frac{2}{5} = \frac{1}{c} \Rightarrow \frac{1}{c} = \frac{16}{95} \Rightarrow c = \frac{95}{16}$$

Putting values in eq.(1)

$$\frac{x}{-5} + \frac{y}{95/18} + \frac{z}{95/16} = 1$$

$$\frac{x}{-5} + \frac{18y}{95} + \frac{16z}{95} = 1$$

Multiplying both sides by 95

$$-19x + 18y + 16z = 95$$

$$19x - 18y - 16z = -95$$

$$19x - 18y - 16z + 95 = 0$$

Q#17: Show that the distance of the point $P(3, -4, 5)$ from the plane $2x + 5y - 6z = 16$ measured parallel to the line

$$\frac{x}{2} = \frac{y}{1} = \frac{z}{-2} \quad \text{is } \frac{60}{7}.$$

Solution:

As the line through $P(3, -4, 5)$ is parallel to given line.

So direction ratios of the line through P are 2, 1, -2.

Hence equation of line through $P(3, -4, 5)$ & parallel to given line is $\frac{x-3}{2} = \frac{y+4}{1} = \frac{z-5}{-2} = t$ (say)

$$\begin{cases} x = 3 + 2t \\ y = -4 + t \\ z = 5 - 2t \end{cases}$$

Any point on this line is $Q(3 + 2t, -4 + t, 5 - 2t)$.

If $Q(3 + 2t, -4 + t, 5 - 2t)$ lies on plane $2x + 5y - 6z = 16$, Then it satisfies the eq. of plane.

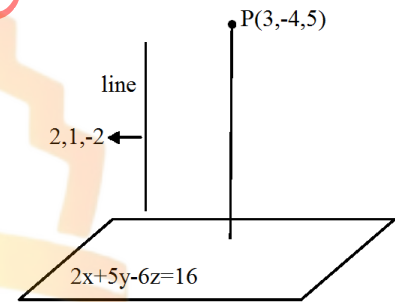
$$2(3 + 2t) + 5(-4 + t) - 6(5 - 2t) = 16$$

$$\Rightarrow 6 + 4t - 20 + 5t - 30 + 12t = 16 \quad \Rightarrow 21t - 44 = 16 \quad t = \frac{20}{7}$$

So coordinates of point are $Q\left(3 + 2\left(\frac{20}{7}\right), -4 + \frac{20}{7}, 5 - 2\left(\frac{20}{7}\right)\right)$

$$= Q\left(\frac{21+40}{7}, \frac{-28+20}{7}, \frac{35-40}{7}\right)$$

$$= Q\left(\frac{61}{7}, \frac{-8}{7}, \frac{-5}{7}\right)$$



Now the required distance

$$= |PQ| = \sqrt{\left(\frac{61}{7} - 3\right)^2 + \left(\frac{-8}{7} + 4\right)^2 + \left(\frac{-5}{7} - 5\right)^2}$$

$$= \sqrt{\left(\frac{61-21}{7}\right)^2 + \left(\frac{-8+28}{7}\right)^2 + \left(\frac{-5-35}{7}\right)^2}$$

$$= \sqrt{\left(\frac{40}{7}\right)^2 + \left(\frac{20}{7}\right)^2 + \left(\frac{-40}{7}\right)^2}$$

$$= \sqrt{\frac{1600}{49} + \frac{400}{49} + \frac{1600}{49}}$$

$$= \sqrt{\frac{1600+400+1600}{49}}$$

$$= \sqrt{\frac{3600}{49}}$$

$$\text{distance} = \frac{60}{7}$$

Q#18: Show that the lines

$$L : x = 3 + 2t, \quad y = 2 + t, \quad z = -2 - 3t$$

$$M : x = -3 + 4s, \quad y = 5 - 4s, \quad z = 6 - 5s \text{ intersect. Find an equation of the plane containing these lines.}$$

Solution:

Given equations of lines are

$$L : x = 3 + 2t, \quad y = 2 + t, \quad z = -2 - 3t$$

$$M : x = -3 + 4s, \quad y = 5 - 4s, \quad z = 6 - 5s$$

Let a point $P(x_0, y_0, z_0)$ is a point of intersection of lines L & M. So this point will lie on both lines

$$L = \begin{cases} x_0 = 3 + 2t \\ y_0 = 2 + t \\ z_0 = -2 - 3t \end{cases} \quad \& \quad M = \begin{cases} x_0 = -3 + 4s \\ y_0 = 5 - 4s \\ z_0 = 6 - 5s \end{cases}$$

Comparing L & M

$$3 + 2t = -3 + 4s \Rightarrow 2t - 4s = -6 \quad \text{----- (1)}$$

$$2 + t = 5 - 4s \Rightarrow t + 4s = 3 \quad \text{----- (2)}$$

$$-2 - 3t = 6 - 5s \Rightarrow -3t + 5s = 8 \quad \text{----- (3)}$$

$$\text{Adding eq. (1) \& (2)} \quad 3t = -3 \Rightarrow t = -1$$

Putting in eq. (1)

$$2(-1) - 4s = -6 \Rightarrow -2 - 4s = -6 \Rightarrow -4s = -4 \Rightarrow s = 1$$

$$\text{Using value of } t \& s \text{ in eq. (3)} \quad -3(-1) + 5(1) = 8 \Rightarrow 8 = 8$$

We see that these values of t & s satisfy eq. (3) So given lines intersect.

Now we find eq. of plane containing given lines L & M

Equations of lines L & M in symmetric form are

$$L : \frac{x-3}{2} = \frac{y-2}{1} = \frac{z+2}{-3} \quad \& \quad M : \frac{x+3}{4} = \frac{y-5}{-4} = \frac{z-6}{-5}$$

As the required plane contains both lines, So it could contain every point of both lines.

A point on the line L is $(3, 2, -2)$

If a, b, c are direction ratios of required plane then eq. of plane through $(3, 2, -2)$ is

$$a(x - 3) + b(y - 2) + c(z + 2) = 0 \quad \text{----- (A)}$$

As this plane contains both lines, So the normal vector of the plane is perpendicular to both the lines

$$\text{Hence } \begin{cases} 2a + 2b - 3c = 0 \\ 4a - 4b - 5c = 0 \end{cases}$$

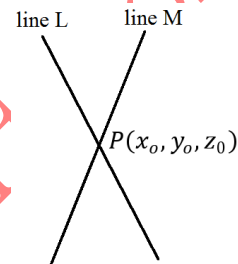
$$\frac{a}{-5-12} = \frac{-b}{-10+12} = \frac{c}{-8-4} \Rightarrow \frac{a}{-17} = \frac{-b}{-2} = \frac{c}{-12} \Rightarrow \frac{a}{17} = \frac{b}{2} = \frac{c}{12}$$

Putting these values of a, b, c in eq. (A)

$$17(x - 3) + 2(y - 2) + 12(z + 2) = 0$$

$$17x - 51 + 2y - 4 + 12z + 24 = 0$$

$$17x + 2y + 12z - 31 = 0 \text{ is required equation of plane.}$$



Q#19: If a, b, c are the intercepts of a plane on the coordinate axes and r is the distance of the origin from the plane, prove that $\frac{1}{r^2} = \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}$.

Solution: Let equation of plane in slope intercept form is

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1 \quad \Rightarrow \quad \frac{1}{a}x + \frac{1}{b}y + \frac{1}{c}z - 1 = 0 \quad \text{--- (A)}$$

Now r is the distance between Origin and plane then By using formula, we get

$$r = \frac{\left| \frac{1}{a}(0) + \frac{1}{b}(0) + \frac{1}{c}(0) - 1 \right|}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}} = \frac{|-1|}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}} \Rightarrow r = \frac{1}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}}$$

Inverting on both sides $\Rightarrow \frac{1}{r} = \sqrt{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}$

Now squaring on both sides $\frac{1}{r^2} = \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}$ hence proved.

Q#20: Find equations of two planes whose distances from the origin are 3 units each and which are perpendicular to the line through the point $A(7, 3, 1)$ and $B(6, 4, -1)$.

Solution: Let equation of set of planes in normal form is $lx + my + nz = p$

As given $p = 3$ units (Distance between plane & Origin)

$$lx + my + nz = \pm 3 \quad \text{--- (1)}$$

Given points are $A(7, 3, 1)$ and $B(6, 4, -1)$.

$$\vec{u} = \overrightarrow{AB} = B(6, 4, -1) - A(7, 3, 1) = -\hat{i} + \hat{j} - 2\hat{k}$$

Here $\overrightarrow{AB}$ is also normal vector of both required planes

$$|\vec{u}| = \sqrt{(-1)^2 + 1^2 + (-2)^2} = \sqrt{6}$$

Now we can find direction cosines of $\overrightarrow{AB}$

$$l = -\frac{1}{\sqrt{6}}, m = \frac{1}{\sqrt{6}}, n = -\frac{2}{\sqrt{6}}$$

Now using values of l, m, n in eq. (1)

$$-\frac{1}{\sqrt{6}}x + \frac{1}{\sqrt{6}}y - \frac{2}{\sqrt{6}}z = \pm 3 \Rightarrow -x + y - 2z = \pm 3\sqrt{6} \quad \text{are required planes.}$$

Checked by: Sir Hameed ullah (hameedmath2017 @ gmail.com)

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Hameed Ullah sb (MSc Mathematics)

